University Of Engineering and Technology, Lahore

Department of Cyber Security

Course Name: Programming Fundamentals	**Course Code: CMPE-112L**
Student Name: Hafiz Obaid Ahmed	**Roll No: 2026(s)-CYS-47**
Semester: Spring 2026	**Session: 2026(s)**
Lab/Project/Assignment #: 14	**CLOs to be covered: CLO2**
Lab Title: OS Library and if __name__ == "__main__"	**Teacher Name: Hafiz Muhammad Mubashar**

Lab Evaluation:

CLO2	Apply appropriate programming techniques to create executable programs to solve well defined problems					
	Level1	**Level2**	**Level3**	**Level4**	**Level5**	**Level6**
(5)						
					Total	/5

Rubrics for Current Lab Evaluation

Scale	**Marks**	**Level**	**Rubric**
Excellent	5	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	4	L2	Submitted the lab tasks but have good understanding
Good	3	L3	Submitted the lab tasks but have weak understanding.
Basic	2	L4	Submitted the lab tasks but have no

			understanding.
Barely Acceptable	1	L5	Submitted only one lab task.
Not Acceptable	0	L6	Did not attempt

LAB No 14: OS Library and if __name__ == "__main__"

Lab Goals/Objectives:

- Understand the purpose of the special variable __name__ in a Python script.
- Learn how the if __name__ == "__main__": construct controls which code runs when a file is executed directly versus imported as a module.
- Practice defining multiple functions in a single file and controlling their execution.

Equipment Required:

- Personal Computer / Laptop
- Python 3.x
- Text Editor / IDE (PyCharm)

Theory:

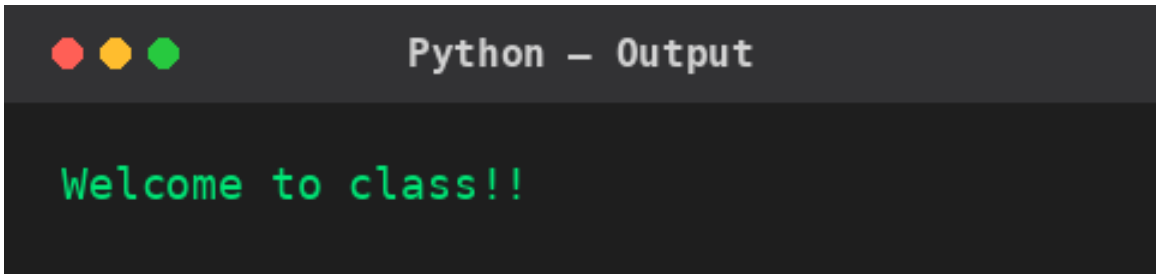
Every Python module has a built-in variable called __name__. When a file is run directly, Python sets __name__ to "__main__"; when the same file is imported into another script, __name__ is instead set to the module's name. The construct if __name__ == "__main__": is used to write code that should only execute when the file is run directly, and not when it is imported elsewhere — this is especially useful for keeping helper functions in a file reusable while still allowing the file to be tested or run standalone. This is closely related to Python's os library, which is commonly used alongside this pattern to write scripts that behave differently depending on how they are invoked.

Lab Work:

Task 1:

```
def display_message():
    print("Welcome to class!!")
def display_message_two():
    print("Welcome to class2!!")
if __name__ == "__main__":
    display_message()
```

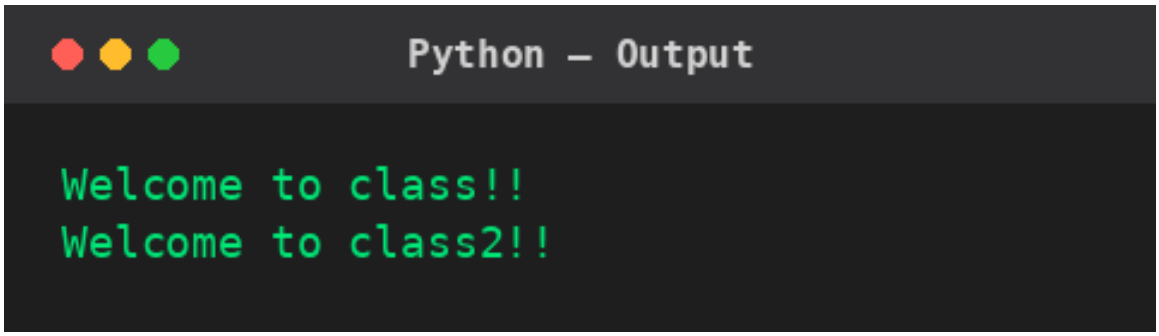
Result:



Task 2:

```
def show_intro():
    print("Welcome to class!!")
def show_intro_two():
    print("Welcome to class2!!")
if __name__ == "__main__":
    show_intro()
    show_intro_two()
```

Result:



Conclusions:

This lab demonstrated the use of the if __name__ == "__main__": construct to control which functions execute when a Python file is run directly, using two simple scripts each defining multiple functions and selectively calling them from the main guard.